



# Uptake mechanism, subcellular distribution, and uptake process of perfluorooctanoic acid and perfluorooctane sulfonic acid by wetland plant *Alisma orientale*

Tuan-Tuan Wang<sup>a,b,c</sup>, Guang-Guo Ying<sup>a,\*</sup>, Liang-Ying He<sup>a</sup>, You-Sheng Liu<sup>a</sup>, Jian-Liang Zhao<sup>a</sup>

<sup>a</sup> The Environmental Research Institute, Guangdong Provincial Key Laboratory of Chemical Pollution and Environmental Safety & MOE Key Laboratory of Theoretical Chemistry of Environment, South China Normal University, Guangzhou 510006, China

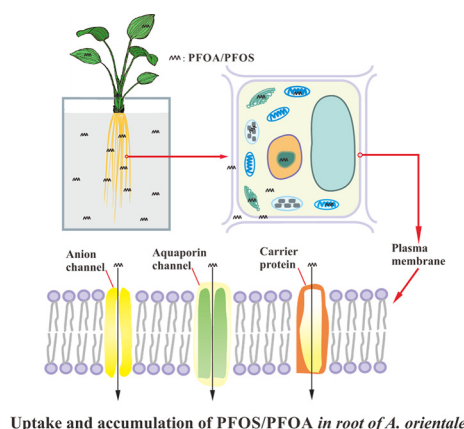
<sup>b</sup> State Key Laboratory of Organic Geochemistry, Guangzhou Institute of Geochemistry, Chinese Academy of Sciences, Guangzhou 510640, China

<sup>c</sup> University of Chinese Academy of Sciences, Beijing 100049, China

## HIGHLIGHTS

- Uptake of PFASs was an active process facilitated by transporters.
- PFOA (PFOS) was mainly distributed in soluble components (cell walls).
- Most of PFASs accumulated in root sub-cellular components were reversible.
- PFASs in soluble fraction could be adsorbed by cell walls and bind with organelles.
- PFASs could be transported through both symplastic and apoplastic routes.

## GRAPHICAL ABSTRACT



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## ABSTRACT

Perfluoroalkyl substances (PFASs) are of particular environmental concern due to their environmental persistence and potential toxicity. Phytoremediation may be used to remove PFASs from wastewater. Here we investigated the uptake mechanism, subcellular distribution, and uptake process of perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate acid (PFOS) in the wetland plant *Alisma orientale* by using a series of hydroponic experiments. Active uptake facilitated by water transporters and anion channels was involved in the uptake of PFASs by plant roots. PFOA and PFOS were mainly distributed in the water-soluble fraction (46.2–70.8%) and in cell walls (45.6–58.4%), respectively. The uptake process was proposed as follows: PFOS and PFOA were first distributed in the soluble fraction; a proportion of PFOS and PFOA were adsorbed gradually by the cell wall, and a proportion of PFOS and PFOA in the cell wall passed through the cell wall and plasmalemma and bind with organelles. PFOS and PFOA were transported from the external solution to the vascular bundle of the plant root through both symplastic and apoplastic routes.

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\* Corresponding author at: No. 378 Outer Ring West Road, Guangzhou University Town, Panyu District, Guangzhou, China.

E-mail address: [guangguo.ying@m.scnu.edu.cn](mailto:guangguo.ying@m.scnu.edu.cn) (G.-G. Ying).

## 1. Introduction

Perfluoroalkyl substances (PFASs) are a class of human-made organic chemicals that have been widely used as surfactants in various industrial and commercial fields (e.g., nonstick food packaging, stain repellents, surfactants and surface protectors) (Buck et al., 2011; Liu et al., 2019). However, due to their chemical and thermal stability and low biodegradability (Giesy and Kannan, 2002), PFASs are found ubiquitously in surface water and aquatic organisms (Pan et al., 2014; Pan et al., 2013). Perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS) are the most representative PFAS chemicals in the rivers of the Pearl River Delta region of South China due to their high concentrations and detection frequencies (Pan et al., 2014). Due to their environmental persistence (Rahman et al., 2014), potential accumulation and toxicity (Ahrens and Bundschuh, 2014), it is necessary to develop an effective technology to remove PFASs from surface water.

Phytoremediation has been widely used to remove persistent organic pollutants from water due to its low cost and environmental friendliness (Reinhold et al., 2010; Sun et al., 2010). The uptake and translocation of PFASs have previously been reported in selected edible crops, vegetables, grasses, and trees but rarely in wetland plants (Blaine et al., 2013; Felizeter et al., 2012; Gobelius et al., 2017; Stahl et al., 2009). The concentrations of PFOA and PFOS in plants were as high as 7.52 mg/kg dw and 254 mg/kg dw (Ghisi et al., 2019), respectively, indicating their high accumulation potential. However, most studies have focused on the effects of chemicals characteristics (e.g., C—F chain length, function group), exposure conditions (e.g., soil organic matter content, soil properties, PFASs concentrations, exposure time, temperature, salinity), and plant species on the uptake, accumulation, and translocation profile of PFASs in plants (Blaine et al., 2013; Blaine et al., 2014a; Blaine et al., 2014b; Felizeter et al., 2014; Felizeter et al., 2012; Krippner et al., 2014; Krippner et al., 2015; Stahl et al., 2009; Xiang et al., 2018; Zhao et al., 2013; Zhao et al., 2016b; Zhao et al., 2016c). The uptake mechanism, subcellular distribution, and uptake process of PFASs in plants still need further research. Compared to terrestrial plants, wetland species show wider and more practical applications for removing inorganic and organic contaminants due to their specific physiological properties (e.g., radial oxygen loss, advanced aerenchyma) (Bhatia and Goyal, 2013). *Alisma orientale* is a typical wetland plant used for phytoremediation and has accumulation potential for pollutants (e.g., nitrogen, phosphorus) (Jiang et al., 2011). In addition, our laboratory previously studied the uptake, accumulation, and translocation of PFOA and PFOS in eight common wetland plants (Wang et al., 2020). It was found that *A. orientale* showed better accumulation and transport capacities for PFOS and PFOA than the other tested plants. Exploring the uptake mechanism, subcellular distribution, and uptake process of PFOS and PFOA in *A. orientale* would provide a theoretical basis for further application of *A. orientale* in phytoremediation.

Plants can take up chemicals through active and/or passive processes (Collins et al., 2006). Active uptake is an energy-consuming process mediated by carrier proteins (e.g., water channels, anion channels), while passive uptake is a process that follows the potential energy gradient, driven by mass flow or diffusion (Zhan et al., 2010). Some passive processes (e.g., simple passive diffusion) are carrier-independent and are similar to the diffusion of chemicals through a semipermeable membrane; these processes can be described by Fick's 1st law (Briggs et al., 1982; Chiou et al., 2001; Trapp, 2000). Other processes are facilitated passive transport, which is mediated by carriers without energy consumption (Wang et al., 2011; Zhan et al., 2012). In general, the uptake of nonionized organic compounds in plant roots is regarded as a simple passive diffusion process (Chiou et al., 2001; Li et al., 2010). Some studies have reported that active uptake and passive transport are involved in the uptake of PFASs in terrestrial crops (e.g., maize and wheat)

(Blaine et al., 2014b; Wen et al., 2013; Zhang et al., 2019). However, few studies have reported on the uptake mechanism of PFASs in wetland plants.

Studying the accumulation of organic pollutants in plant cell organelles is an appropriate method of demonstrating the intracellular accumulation capacity of organic pollutants. Exploring the subcellular distribution of organic pollutants in plants is useful in understanding plant tolerances to pollutants. For example, binding with cell wall and vacuolar sequestration play important roles in plant tolerances to organic pollutants (Wild et al., 2005). High accumulation of organic pollutants in organelles may result in the changes in organelle structure and function. It has been reported that neutral lipophilic organic pollutants (e.g., phenanthrene and pyrene) mainly accumulate in cell walls and cell organelles (Kang et al., 2010; Wild et al., 2004). However, knowledge about the subcellular distribution of anionic PFASs, which are hydrophobic and oleophobic, is still limited. Directly “seeing” chemicals in plant roots would enhance our mechanistic and process-based understanding of environmentally relevant chemicals (Wild et al., 2005). Transmission electron microscopy equipped with energy dispersive spectroscopy (TEM-EDS) has been used to observe the cell-level distribution of heavy metals (e.g., Zn, and Cd) in plant roots (Wu et al., 2015). It may also be useful for observing the cell-level distribution of PFASs in plant roots based on the identification of the element F, which is not a constituent element of plants.

The objective of this study was to explore the uptake mechanism, subcellular distribution, and uptake process of PFOS and PFOA in the wetland plant *A. orientale* by using a series of hydroponic experiments. TEM-EDS was applied to directly visualize the cell-level distribution of PFOS and PFOA in plant roots to understand the uptake process. The results from this study may help reveal how wetland plants take up and distribute PFOS and PFOA from macro- and micro perspectives.

## 2. Materials and methods

### 2.1. Standards, reagents and materials

The physicochemical properties of PFOA and PFOS are shown in Table S1. Isotope-labeled standards MPFOA ( $^{13}\text{C}_4$ -PFOA) and MPFOS ( $^{13}\text{C}_4$ -PFOS) were used as internal standards for the quantification. 2,4-Dinitrophenol (2,4-DNP), glycerol, silver nitrate ( $\text{AgNO}_3$ ), anthracene-9-carboxylic acid (9-AC), 4,4-diisothiocyanostibene-2,2-disulfonate (DIDS), and 5-nitro 2-(3-phenylpropylamine) benzoic acid (NPPB) were used as inhibitors. HPLC-grade methanol and Milli-Q water were used throughout the study. Florisil solid phase extraction (SPE) cartridges and Bond Carbon-GCB (10 g, 120–400 mesh size) were used in the extraction cleanup. All reagents and materials and their suppliers are shown in the Supplementary Information (SI Text S1).

### 2.2. Plant preparation

*A. orientale* seedlings with height of approximately 10 cm were obtained from a nursery in Foshan, China. The soil was washed off the plant roots with running water, and the plants were transferred to a 1/2 strength Hoagland nutrient solution (Table S2) for two weeks of acclimatization. Then, the plants were transferred to a 1/2 strength Hoagland nutrient solution spiked with PFOA or PFOS in subsequent experiments. The hydroponic experiments were performed in a 1.2 L PP plastic bucket covered with a black shading bag. The plants were fixed in the holes of the barrel cover, which were filled with a sponge to support the plants. Each bucket, which contained two plants, was filled with 1.0 L nutrient solution. Only the roots of plants were exposed to the nutrient solution. The water transpired by plants was measured by weight and supplemented with the nonspiked nutrient solution.

### 2.3. Exposure experiments

#### 2.3.1. Experiment 1: time-dependent uptake and transport of PFOS and PFOA

Wetland plants with uniform height were transferred to 1/2 strength Hoagland solution containing 1.0 mg/L PFOS or 1.0 mg/L PFOA. The plant roots and shoots were sampled at increasing exposure time intervals (0, 2, 4, 8, 12, 24, 48, 72, 96, and 120 h), rinsed thoroughly with running water and Milli-Q water three times and blotted dry with tissue paper. Then, the PFOS and PFOA in the roots and shoots were extracted and determined. Each treatment had three replicates.

#### 2.3.2. Experiment 2: concentration-dependent uptake of PFOS and PFOA

Plants with uniform height were transferred to 1/2 strength Hoagland solution spiked with PFOS or PFOA at concentrations of 0.0, 100.0, 200.0, 400.0, 500.0, 800.0, and 1000.0  $\mu\text{g/L}$  (pH = 6) for 8 h. Then, the plant roots were sampled, rinsed thoroughly with running water and Milli-Q water three times and blotted dry with tissue paper. The PFOS and PFOA in roots were extracted and determined. Each treatment had three replicates.

#### 2.3.3. Experiment 3: effect of metabolic, aquaporin and anion-channel inhibitors on uptake

The concentration of PFOS or PFOA was set at 1.0 mg/L, and an inhibitor was added. 2,4-DNP was used as a metabolic inhibitor at a concentration of 0.5 mmol/L. For the aquaporin inhibitors, 0.5 mmol/L glycerol and 0.5  $\mu\text{mol/L}$   $\text{AgNO}_3$  were selected; 10  $\mu\text{mol/L}$  anthracene-9-carboxylic acid (9-AC), 5  $\mu\text{mol/L}$  4,4-diisothio-cyanostibene-2,2-disulfonate (DIDS) and 5  $\mu\text{mol/L}$  5-nitro 2-(3-phenylpropylamine) benzoic acid (NPPB) were applied as anion channel inhibitors. Plants grown in PFOS- or PFOA-spiked nutrient solution without inhibitors were used as controls. After exposure for 8 h, the plant roots were sampled, rinsed thoroughly with running water and Milli-Q water three times, and blotted dry with tissue paper. Then, the PFOS and PFOA in roots were extracted and determined. Each treatment had three replicates.

#### 2.3.4. Experiment 4: uptake competition between PFOS and PFOA

Uniform plants were planted in 1/2 strength Hoagland solution spiked with 1.0 mg/L PFOS, 1.0 mg/L PFOA, or a mixture of 1.0 mg/L PFOS and 1.0 mg/L PFOA. After 8 h of exposure, the plant roots were sampled, rinsed thoroughly with running water and Milli-Q water three times, and blotted dry with tissue paper. Then, the PFOS and PFOA in roots were extracted and determined. Each treatment had three replicates.

#### 2.3.5. Experiment 5: subcellular distribution of PFOS and PFOA during the uptake

Plants with uniform height were transferred to 1/2 strength Hoagland solution containing 1.0 mg/L PFOS or 1.0 mg/L PFOA. The plants were sampled at increasing exposure time intervals (0, 0.5, 1, 2, 4, 8, and 16 d). The plants were rinsed thoroughly with running water and Milli-Q water three times, blotted dry with tissue paper, and divided into roots, stems, and leaves. Then, the PFOS and PFOA in the subcellular components (e.g., cell walls, water-soluble fraction, and cell organelles) of the roots, stems, and leaves were determined. Each treatment had three replicates. The solution in each replicate was exchanged once every 5 d to keep the spiked concentration at a constant level and to avoid microbial growth in the nutrient solution.

#### 2.3.6. Experiment 6: the dissipation of PFOS and PFOA in plant roots

The plants that had been incubated for 72 h in 1/2 strength Hoagland solution containing 1.0 mg/L PFOS or 1.0 mg/L PFOA were removed from the PFOS or PFOA spiked solution. The plant roots were rinsed thoroughly with Milli-Q water three times. The plants were then transplanted into 1/2 strength Hoagland solution without PFOA and PFOS. The plants and hydroponic solution were sampled at increasing

dissipation time intervals (0, 2, 4, 8, and 16 d). The plants were rinsed thoroughly with running water and Milli-Q water three times, blotted dry with tissue paper, and divided into roots and shoots. Then, the PFOS and PFOA in the roots, shoots, and subcellular components (e.g., cell walls, soluble fraction, and organelles) of the roots and in the hydroponic solution were extracted and determined. Each treatment had three replicates. The solution in each replicate was exchanged at each sampling interval (2, 4, 8, and 16 d).

### 2.4. Extraction and determination

The subcellular components were extracted based on previous methods with slight modifications (Wang et al., 2019; Zhao et al., 2015). In brief, fresh plant tissues (e.g., roots, stems, leaves) (1.0 g) were homogenized in precooled extraction buffer solution (50 mM Tris (hydroxymethyl)aminomethane (Tris-HCl), 250 mM sucrose, 1.0 mM dithioerythritol, and 5.0 mM ascorbic acid and adjusted to pH 7.5 with 1.0 M NaOH) with a chilled mortar and pestle. The homogenate was transferred to a 50 mL tube and centrifuged at 300g for 30 s to obtain the cell wall fraction from the precipitate. The supernatant solution was centrifuged again at 20,000g for 45 min to sediment the organelles (including all organelles except the vacuole). The supernatant solution was taken as the water-soluble fraction including the vacuole. All steps were performed at 4 °C. The cell walls and organelles were vacuum freeze-dried and extracted as root and stem tissues.

The extraction of PFOS and PFOA from plant tissues (roots, stems, leaves, cell walls, and organelles) was performed based on previous methods with slight modifications (Felizeter et al., 2012; Ullah et al., 2012). Further details are available in SI Text S2. The PFASs in the subcellular soluble fraction were mixed with chromatographic pure methanol (v/v = 50:50) containing internal standards, and then determined directly. Water samples (e.g., the hydroponic solution in the dissipation experiment) were extracted by solid phase extraction (SPE) based on our previous methods (Pan et al., 2014). Further details are available in SI Text S3. The target PFOA and PFOS in the extracts were analyzed by an Agilent 1200 HPLC system coupled with an Agilent 6460 Triple Quadrupole (QQQ) mass spectrometer (Agilent Technologies, Santa Clara, USA) that was operated in electrospray negative ionization mode. The detailed operating conditions can be found in our previous study (Pan et al., 2014).

### 2.5. Transmission electron microscopy analysis

Plants that were cultured in the PFAS-spiked solution (1.0 mg/L PFOS or 1.0 mg/L PFOA) for 72 h were set as the treatment group. Plants that were cultured in the solution not spiked with PFOS or PFOA for 72 h were set as the blank group. The PFOS and PFOA in the plant roots of the treatment and blank groups were identified with field emission TEM-EDS. Fresh root sections in the root hair zone were cut into 2 mm pieces, fixed in 2.5% (v/v) glutaraldehyde and 1% (v/v) osmium tetroxide successively, dehydrated using graded ethanol (30%–100%, v/v), embedded in Epon-812 resin, and sliced with a Leica UCT ultramicrotome. The sections (50–100 nm) were mounted on copper grids, stained with 4% (w/v) uranyl acetate and lead citrate and then observed under transmission electron microscopy (JEM-1400 Plus, Hitachi, Tokyo, Japan). The energy spectrum of fluorine was determined using an energy dispersive X-ray spectrometer (IXRF Systems, Houston, TX, USA). We first scanned the root slice surface to determine whether fluorine (F) was present. Then, the points that included the cell wall, the intercellular space and various cell organelles (e.g., nucleus, chondriosome, Golgi apparatus, and cytoplasm) from the epidermis to the xylem vessel were examined to determine whether F was present. PFOS (or PFOA) in the root cross-sections was identified by the existence of the energy spectrum of F, which was not observed in the blank root sample (Fig. S1).

## 2.6. Quality control and method performance

Quality assurance and quality control (QA/QC) procedures were followed during the extraction and analysis. The spiked controls, procedural blanks and recoveries were determined for each set of extractions. The solvent blanks, procedural blanks and standard controls (10 µg/L) were run successively every 12 samples to check for any carryover or background contamination as well as to check the precision and accuracy of the instrument. The recoveries, limits of detection (LOD) and limits of quantification (LOQ) for PFOA and PFOS in the plant and water samples are given in Table S3. The LOD and LOQ were calculated based on the signal-to-noise ratio (SNR) of the blank sample at a low concentration of standards. The LOD and LOQ were defined as 3 times and 10 times the concentration of standards divided by the SNR, respectively. The SNR was calculated by Agilent Masshunter qualitative software. Teflon-coated labware and glass containers were avoided throughout the experiment. Polypropylene (PP) versions of labware were used instead (Ullah et al., 2012).

## 2.7. Data analysis

The concentrations of PFASs in plant tissues (e.g., roots, stems, leaves, subcellular components) in the present study were expressed on a fresh weight basis (fw). The time-dependent uptake and transport of PFASs were fitted to one-compartment first-order kinetics (Wen et al., 2013):

$$C_t = C_0 + \alpha/k_e (1 - e^{-k_e t}) \quad (1)$$

where  $C_t$  is the concentration of PFOS or PFOA (mg/kg fw) in roots and shoots at time  $t$  (h);  $k_e$  is the excretion rate constant (1/h); and  $\alpha$  is the uptake flux constant (mg/(kg·h) fw), which equals  $k_u \cdot C_w$ , where  $k_u$  is the uptake rate constant (L/(kg·h)) and  $C$  is the bioavailable external concentration in solution (mg/L). The concentration of PFOS or PFOA in roots and shoots at the beginning of the experiment,  $C_0$ , was assumed to be a pool of constant magnitude. Subsequently, Eq. (1) was used to calculate the steady-state tissue PFOS or PFOA concentration ( $C_{ss}$ ) in roots and shoots, which equaled the ratio of the uptake and transport flux constant to the elimination constant ( $\alpha/k_e$ ). The steady-state concentration was used to calculate the root concentration factor (the ratio of the PFOS or PFOA concentration in plant roots to the solution concentration, RCF) and the transport factor (the ratio of the concentrations of PFOS or PFOA in shoots to those in roots, TF).

The concentration-dependent uptake of PFOS and PFOA was described with a Michaelis-Menten equation (Wen et al., 2013):

$$V = V_{\max} C / (K_m + C) \quad (2)$$

where  $V$  is the uptake rate (µg/(kg·h) fw);  $C$  is the concentration of PFOS or PFOA in the solution (µg/L);  $K_m$  (µg/L) is the Michaelis constant which equals the substrate concentration giving half the maximum uptake rate, and  $V_{\max}$  (µg/(kg·h) fw) is the maximum uptake rate when all available carrier sites are loaded (Zhan et al., 2018).

If without a special request, all statistical analyses were conducted with the software SPSS 22 for Windows (SPSS Inc., Chicago, IL). Nonlinear regression analyses were conducted by the least-squares method. An independent-sample  $t$ -test was used to assess the significance of the effect of inhibitors and competition on the uptake of PFASs by plant roots. One-way ANOVA was used to assess the significance of the difference among different dissipation times. The data used for one-way ANOVA were normally distributed. A one-sample Kolmogorov-Smirnov test of variates was used to verify the normality of the distribution. The significant difference for the one-compartment first-order kinetics regression coefficients of PFOS and PFOA was performed based on the sum of square reduction test in SAS. The significant difference for the Michaelis-Menten regression coefficients of PFOS and

PFOA in plant roots was performed based on Chow test in SAS. Statements of significant differences are based on  $p < 0.05$ .

## 3. Results

### 3.1. Uptake of PFOS and PFOA by wetland plants

The concentrations of PFOS and PFOA in plant roots (Fig. S2 and Table S4) increased linearly from 0 h to 12 h, then increased slowly from 12 h to 48 h and reached saturation at 120 h. The time-dependent uptake and transport data for PFOS and PFOA fitted well to the kinetic Eq. (1) ( $R^2 > 0.97$ ) (Table S5). The higher  $\alpha$  and lower  $k_e$  of PFOS in roots resulted in a higher RCF value for PFOS than for PFOA. In shoots, the  $\alpha$  and  $k_e$  values of PFOA were 1.6 and 4.6 times higher than those of PFOS, respectively (Table S5). The lower shoot steady-state concentration of PFOA ( $C_{ss}$ ) was due to its much higher  $k_e$  value than that of PFOS. The TF value was calculated by the ratio of root  $C_{ss}$  to shoot  $C_{ss}$  which was equal to  $(\alpha_{\text{shoot}}/\alpha_{\text{root}})/(k_{e-\text{shoot}}/k_{e-\text{root}})$ , where  $\alpha_{\text{shoot}}$ ,  $\alpha_{\text{root}}$ ,  $k_{e-\text{shoot}}$  and  $k_{e-\text{root}}$  are the  $\alpha$  of the shoot,  $\alpha$  of the root,  $k_e$  of the shoot, and  $k_e$  of the root, respectively. The  $\alpha_{\text{shoot}}/\alpha_{\text{root}}$  and  $k_{e-\text{shoot}}/k_{e-\text{root}}$  values of PFOA were 3.5 and 2.9 times higher than those of PFOS, respectively. Thus, the TF of PFOA was 1.15 times higher than that of PFOS (Table S5).

Based on the one-compartment first-order kinetics, the accumulation of PFOS and PFOA in roots was the results of uptake and excretion of PFASs in roots. The slope of the curve ( $\alpha \cdot e^{-k_e t}$ ) in Fig. S2 indicated the accumulation rate of PFOS and PFOA in plant roots. The concentrations of PFOS and PFOA in plant roots increased linearly from 0 h to 12 h, and the accumulation rates were relatively stable and high (Fig. S2). This was due to the high uptake and negligible excretion of PFOS and PFOA in plant roots. Thus, the accumulation of PFOS and PFOA in plant roots was the results of root uptake. As the excretion of PFOS and PFOA in roots increased in 12–48 h, the accumulation rates continuously decreased. The accumulation rates of PFOS and PFOA in 48–120 h were close to 0 due to the equilibrium between root uptake and root excretion (Table S3). Exploring the effects of other factors (e.g., exposure concentration, inhibitors, uptake competition in experiments 2–4) on the uptake of PFOS and PFOA in roots, the effect of root excretion should be minimum and an exposure time <12 h was suitable. To ensure that the concentrations of PFOS and PFOA in roots were high enough to be determined by LC-MS/MS, the exposure time in experiments 2–4 was set as 8 h.

The concentration-dependent uptake of PFOS and PFOA by *A. orientale* roots was nonlinear, and could be adequately described by the Michaelis-Menten function ( $R^2 > 0.99$ ) (Fig. 1 and Table S6), with

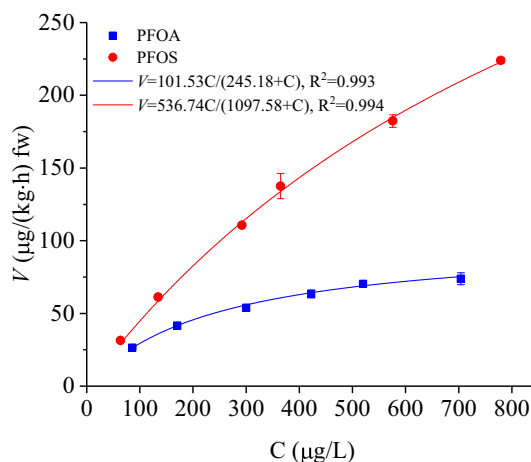


Fig. 1. Concentration-dependent uptake of PFOS and PFOA by plant roots. Lines represent the fits to Eq. (2). The uptake time is 8 h.  $V$ : the uptake rate of PFOS or PFOA (µg/(kg·h) fw);  $C$ : the concentration of PFOS or PFOA in the solution (µg/L).



an apparent  $V_{\max}$  of 101.5  $\mu\text{g}/(\text{kg}\cdot\text{h})$  fw and  $K_M$  of 245.2  $\mu\text{g}/\text{L}$  for PFOS and an apparent  $V_{\max}$  of 536.7  $\mu\text{g}/(\text{kg}\cdot\text{h})$  fw and  $K_M$  of 1097.6  $\mu\text{g}/\text{L}$  for PFOA (Table S7).

The uptake of PFOS and PFOA was reduced by 14.4% and 24.1%, respectively, when the plants were treated with the metabolic inhibitor 2,4-DNP (Fig. 2a and Table S8). The uptake of PFOS and PFOA decreased by 30.9–33.9% and 22.3–25.3% when treated with the aquaporin inhibitors glycerol and  $\text{AgNO}_3$  (Fig. 2a), respectively. The anion channel blocker 9-AC inhibited the uptake of PFOS by 11.1%, and DIDS inhibited the uptake of PFOA by 16.4% (Fig. 2b). The

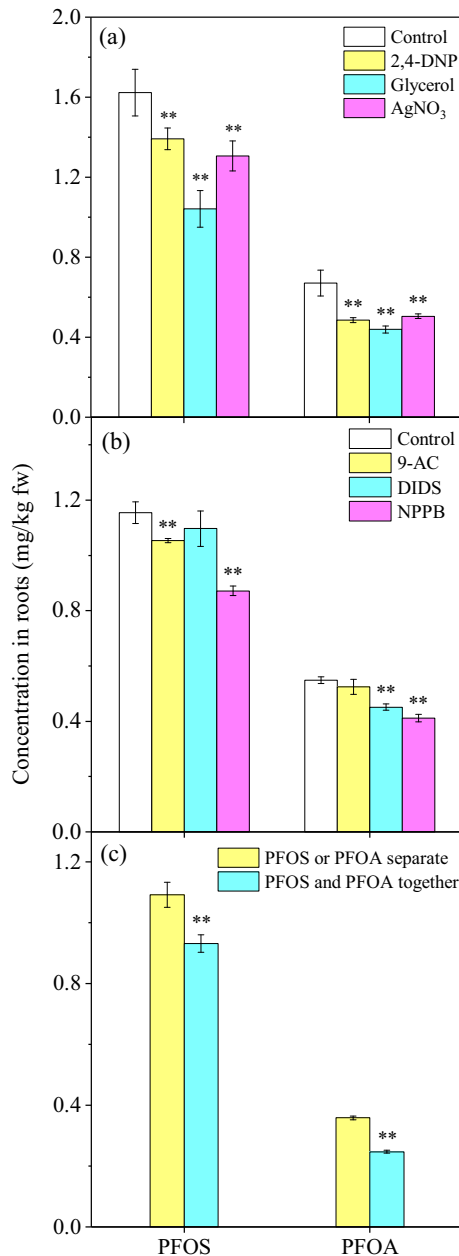
anion channel blocker NPPB inhibited the uptake of PFOS and PFOA by 27.6% and 23.9%, respectively (Fig. 2b). The presence of PFOS inhibited the uptake of PFOA in *A. orientale* roots by 31.0%, and the presence of PFOA inhibited the uptake of PFOS by 14.7% (Fig. 2c). Therefore, a competitive inhibition occurred when PFOS and PFOA were present in the culture solution.

### 3.2. Subcellular distribution of PFASs in plant roots

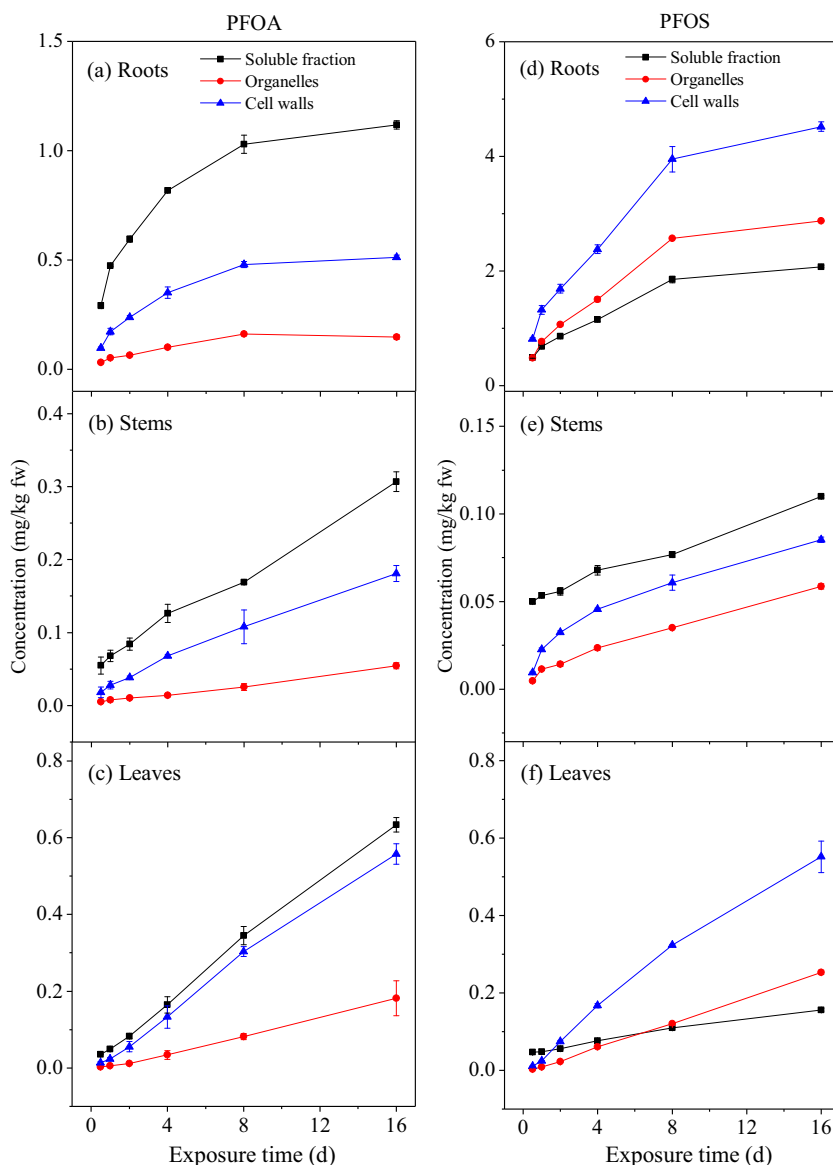
As the exposure time increased from 0.5 d to 16 d, the concentrations of PFOS and PFOA in stems, leaves, and the subcellular components of both stems and leaves increased linearly (Fig. 3 and S3). As the uptake time increased, the concentrations of PFOS and PFOA in roots and in the subcellular components of roots increased continuously from 0.5 d to 16 d. However, the concentrations of PFOS and PFOA in the external solution remained unchanged during the exposure time, and the concentration gradient of PFOS and PFOA between the external solution and in the root subcellular components decreased. Thus, the concentrations of PFOS and PFOA in roots and in the subcellular components of roots increased more slowly from 0.5 d to 16 d (Fig. 3 and S3). The concentrations of PFOS and PFOA in plants from 2 d to 16 d decreased in the following order: roots (0.90–9.47 mg/kg fw) > leaves (0.15–1.37 mg/kg fw) > stems (0.10–0.54 mg/kg fw) (Table S9). This indicated that PFOS and PFOA accumulated more in leaves than in stems. The water-soluble fraction had the highest concentrations of PFOA in roots, stems and leaves, ranging from 0.04–1.12 mg/kg fw, which was 1.1–3.0 and 3.5–12 times higher than those in cell walls (0.01–0.56 mg/kg fw) and organelles (<0.01–0.18 mg/kg fw), respectively (Table S10). Cell walls had the highest concentrations of PFOS in roots and leaves (except in leaves at 0.5–1 d), ranging from 0.07–4.52 mg/kg fw, which was 1.5–3.3 and 1.5–3.5 times that in cell organelles (0.02–2.87 mg/kg fw) and the soluble fraction (0.06–2.08 mg/kg fw) (Table S10). In the stems, the concentration of PFOS in the soluble fraction was highest, in the range of 0.05–0.11 mg/kg fw, which was 1.9–10.4 and 1.3–5.2 times that in cell organelles (<0.01–0.06 mg/kg fw) and cell walls (0.01–0.09 mg/kg fw), respectively (Table S10).

As the exposure time increased from 0.5 d to 8 d, the proportions of PFOA and PFOS in the soluble fraction of roots, stems, and leaves decreased from 69.0–70.8% to 47.2–61.6% and from 27.4–77.7% to 19.8–44.5%, respectively (Fig. 4 and Table S11). Within the same period, the PFOA and PFOS distributed in cell walls increased from 22.2–25.3% to 28.7–40.6% and from 14.8–45.6% to 35.2–58.4%, respectively (Table S11). The proportion of PFOA and PFOS in organelles increased from 5.7–7.7% to 9.7–13.2% and from 5.5–27.0% to 20.3–30.7%, respectively (Table S11). After 8 d of exposure, the distribution of PFOA and PFOS in the subcellular components approached a relatively stable state (Fig. 4). The decrease in PFOS and PFOA in the soluble fraction largely corresponded to the increase of PFOS and PFOA in cell walls and organelles; this suggested that PFOS and PFOA first accumulated in the water-soluble fraction and then were gradually absorbed by cell walls and that a certain proportion absorbed in the cell wall could pass through the cell wall and plasmalemma and bind with organelles.

The PFOA in the soluble fraction (46.2–70.7%) was 1.1–5.2 and 1.9–12 times higher than that in the cell walls (22.8–40.6%) and organelles (5.7–13.2%), respectively (Fig. 4 and Table S11), which indicated that the soluble fraction was the main storage component for PFOA. This was also applicable to PFOS in stems, as most PFOS (43.3–77.7%) accumulated in the soluble fraction in stems (Table S11). In roots and leaves (from 2 d to 16 d), the proportion of PFOS distributed in the cell walls (45.6–58.4%) was higher than that in the organelles (14.8–30.7%) and in the soluble fraction (16.3–36.7%) (Fig. 4 and Table S10). This indicated that in roots and leaves, PFOS was mainly distributed in the cell wall, while it was mainly distributed in the soluble fraction in stems.



**Fig. 2.** Effects of various inhibitors on the root uptake of PFOA and PFOS (a, b) and uptake competitions between PFOS and PFOA (c). Metabolic inhibitor: 2,4-dinitrophenol (2,4-DNP); aquaporin inhibitors: glycerol and silver nitrate ( $\text{AgNO}_3$ ); anion channel inhibitors: anthracene-9-carboxylic acid (9-AC), 4,4-diisothiocyanostibene-2,2-disulfonate (DIDS), and 5-nitro 2-(3-phenylpropylamine) benzoic acid (NPPB). PFOS or PFOA separate: plants were planted in solution spiked with 1.0 mg/L PFOS or 1.0 mg/L PFOA; PFOS and PFOA together: plants were planted in solution spiked with a mixture of 1.0 mg/L PFOS and 1.0 mg/L PFOA. An asterisk indicates a significant difference at  $p < 0.05$ , and two asterisks indicate a significant difference at  $p < 0.01$ .

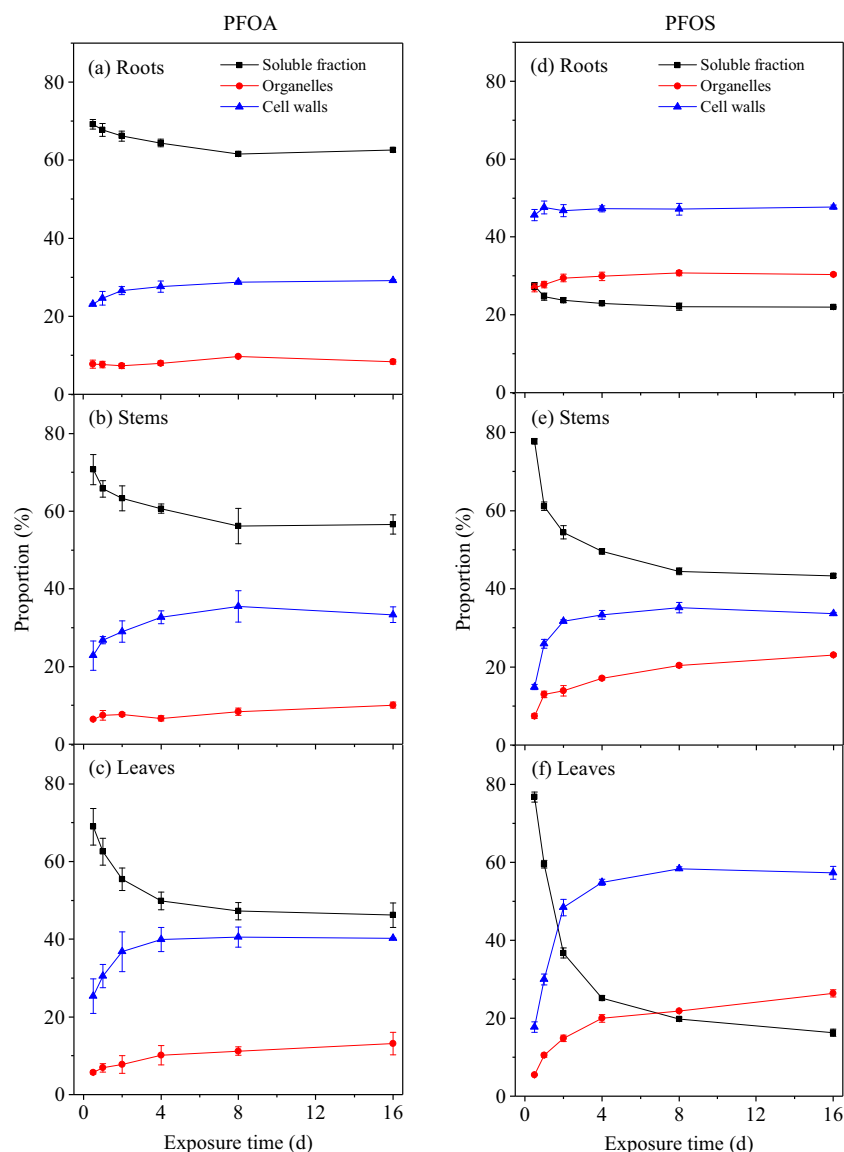


**Fig. 3.** Concentrations of PFOS and PFOA in the subcellular components (e.g., cell walls, organelles, and soluble fraction) of roots, stems, and leaves in the uptake experiment at different exposure times.

### 3.3. The dissipation of PFASs in plant roots

The concentrations of PFOA and PFOS in plant roots decreased from 0.59 to 0.11 mg/kg fw and 2.19 to 0.44 mg/kg fw after transplantation into solutions that were free of PFOA and PFOS, respectively, for 16 d (Fig. 5a and d and Table S12). Accordingly, the concentrations of PFOA and PFOS in the hydroponic solution increased from 0.00 to 2.94 and 0.00 to 10.64  $\mu\text{g/L}$ , respectively, in the 16 d dissipation experiment (Table S12). The concentrations of PFOA and PFOS in plant shoots increased from 0.07 to 0.09 mg/kg fw and from 0.05 to 0.06 mg/kg fw, respectively, from 0 d to 4 d (Fig. 5a and d and Table S12). This indicated that the decrease in the root concentrations of PFOA and PFOS was related to the upward translocation of PFOA and PFOS to shoots (0 to 4 d) and their efflux into the hydroponic solution (Table S12). From 4 d to 16 d, the PFOA and PFOS concentrations in shoots decreased gradually from 0.09 to 0.07 mg/kg fw and from 0.06 to 0.04 mg/kg fw (Fig. 5a and d), respectively, suggesting that the PFOA and PFOS accumulated in shoots could be transported downward to the hydroponic solution.

To determine the contribution of upward translocation and efflux into the solution to the dissipation of PFOS and PFOA in roots, we calculated the mass distribution balance of PFOA and PFOS during the dissipation experiment. The reduced mass loadings of PFOA and PFOS in plant roots were 2.27–3.29  $\mu\text{g}$  and 7.45–12.41  $\mu\text{g}$ , respectively, from 2 d to 16 d (Table S12). The increased mass loadings of PFOA and PFOS in the hydroponic solution were 1.82–2.94  $\mu\text{g}$  and 6.29–10.64  $\mu\text{g}$ , respectively, from 2 to 16 d, which was responsible for 72.9–89.4% and 82.7–85.7% of the reduced mass loadings in roots (Table S12). The increased mass loadings of PFOA and PFOS in shoots were 0.15–0.42  $\mu\text{g}$  and 0.14–0.28  $\mu\text{g}$ , respectively, from 2 to 8 d, which was responsible for 5.0–16.1% and 1.9–3.0% of the reduced mass loading in roots (Table S12). The contribution of the upward translocation to shoots was negligible at 16 d because at 16 d, 0.05  $\mu\text{g}$  PFOA and 0.06  $\mu\text{g}$  PFOS were transported downward from the shoots to the hydroponic solution compared to the amounts at 0 d (Table S12). Other processes (e.g., adsorption to surfaces of the system) also played a certain role (11.0–12.0% for PFOA and 13.6–14.8% for PFOS) in the reduced mass loadings of PFASs in roots (Table S12).

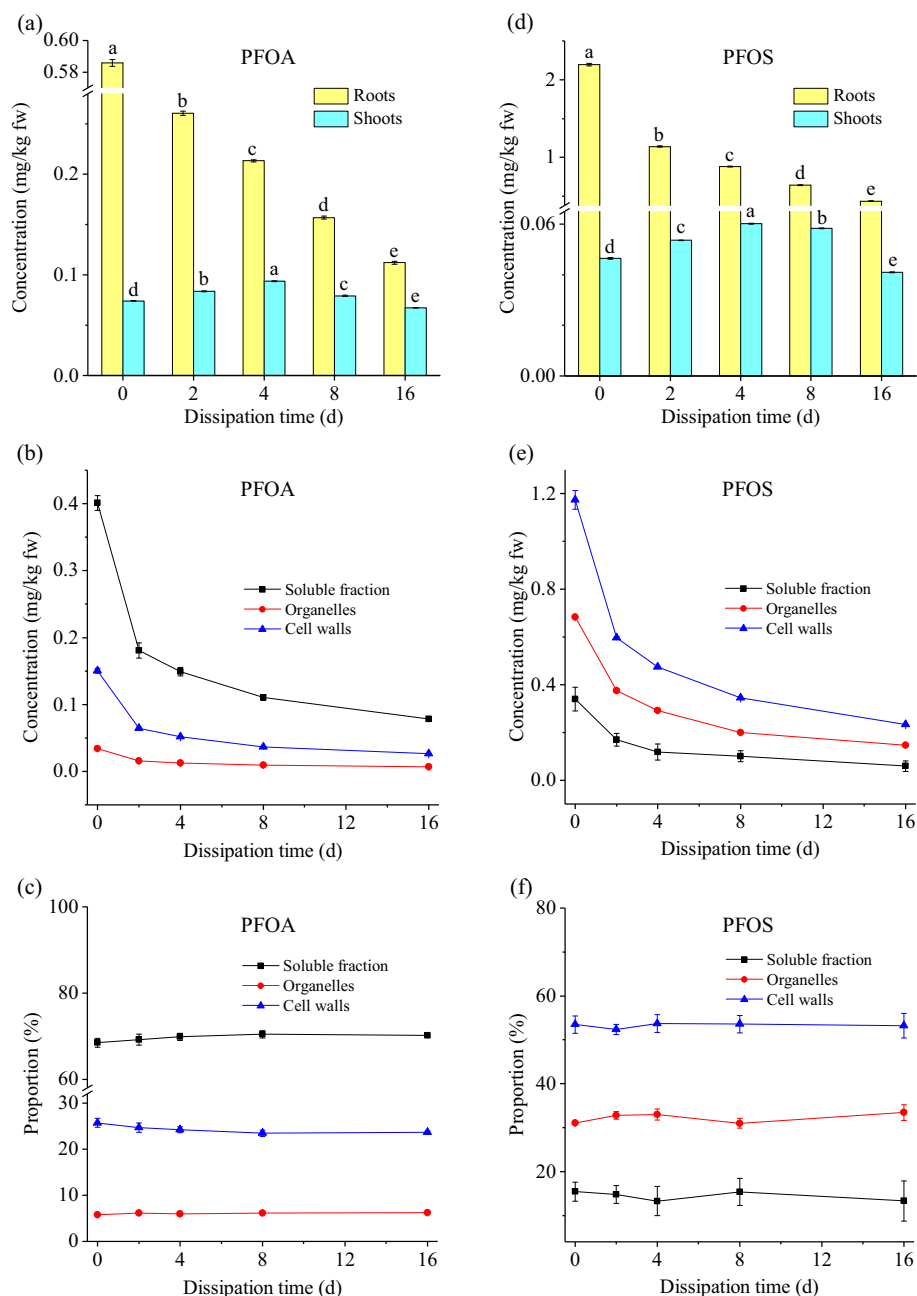


**Fig. 4.** Proportions of PFOS and PFOA in the subcellular components (e.g., cell walls, organelles, and soluble fraction) of roots, stems, and leaves in the uptake experiment at different exposure times.

To ascertain the location of PFOS and PFOA efflux in roots, the concentration of PFOS and PFOA in the subcellular components were also determined. As the dissipation period lengthened, the concentrations of PFOS and PFOA in the water-solution fraction, cell walls, and organelles decreased gradually from 0 to 16 d (Fig. 5 b and e). Approximately 80.4–82.6%, 80.2–82.4%, and 78.6–79.6% of the PFASs accumulated in solution fraction, cell walls, and organelles, respectively, were discharged into the culture solution when the plants were transplanted into a hydroponic solution free of PFASs for 16 d (Table S13). During the dissipation period, the proportion of PFOS and PFOA in the three subcellular components remained stable (Fig. 5 c and f). The proportions of PFOS in the cell walls, cell organelles, and soluble fraction were approximately 52.4–53.7%, 31.0–33.4% and 13.3–15.4% of PFOS, respectively, and they were 23.4–25.7%, 5.82–6.21%, and 68.5–70.4% of PFOA, respectively (Fig. 5 c and f). This indicated that PFOS and PFOA were distributed mainly in the cell walls and soluble fraction, respectively, which was consistent with the subcellular distribution of PFOS and PFOA observed in the uptake experiment (Fig. 4).

### 3.4. Cell-level distribution visualization

PFOS and PFOA were found in cell walls, intercellular spaces, and various organelles (e.g., cell nucleus, mitochondria, chloroplast, and cytoplasm) (Fig. S4). This suggested that PFOS and PFOA in extracellular solution could bind with cell walls and that a proportion of the PFOS and PFOA in cell walls could pass through the cell wall and plasma-lemma and bind with organelles. This was consistent with the uptake process of PFASs in roots obtained from the subcellular distribution results (Figs. 3 and 4). In the cortex of *A. orientale*, PFOS and PFOA were found in both apoplastic space (e.g., the cell wall and intercellular spaces) and symplastic space (e.g., plant endophytes) (Fig. S4). Similar results were also found in the vascular bundle, in which PFOA and PFOS were found in both symplastic space (e.g., the cell nucleus, mitochondria, chloroplast, and cytoplasm) and apoplastic space (e.g., the cell walls and intercellular space) (Fig. S4). In addition, the mass percentage and atomic percentage of fluorine, and the relative mass percentage and atomic percentage of fluorine compared to the most abundant element, carbon, are shown in Table S14.



**Fig. 5.** Concentrations of PFOS and PFOA in plant tissues (e.g., roots and shoots) (a and d) and subcellular fractions (e.g., cell walls, organelles, and soluble fraction) of roots (b and e) in the dissipation experiment at different dissipation times. Proportions of PFOS and PFOA in subcellular fractions (e.g., cell walls, organelles, and soluble fraction) of roots as a function of dissipation time (c and f). Letters (a, b, c, d, e) indicate significant differences in concentrations among different dissipation times ( $p < 0.05$ ).

## 4. Discussion

### 4.1. Uptake mechanism of PFOS and PFOA

Plants can take up chemicals via non-facilitated diffusion and/or through carriers in cell membranes. For instance, some herbicides (e.g., glyphosate) can be absorbed by broad bean (*Vicia faba*) via a phosphate transporter in the plasma membrane (Denis and Delrot, 1993), while atrazine and phenol are taken up in rice seedlings cells without transporters (Su and Zhu, 2005). When the concentration-dependent uptake kinetics follow a Michaelis-Menten pattern, it indicates uptake via a carrier-mediated process (Zhan et al., 2010). In the present study, the concentration dependent uptake of PFOS and PFOA by *A. orientale* roots followed the Michaelis-Menten model well (Fig. 1),

indicating that the uptake of PFOS and PFOA was consistent with a carrier-mediated process. Similar results were also found in maize and wheat (Wen et al., 2013; Zhang et al., 2019).

Carrier-mediated processes include active uptake and passive transport (Wang et al., 2011). To further distinguish active uptake from facilitated passive transport, a metabolic inhibitor was used to inhibit active uptake by disabling cellular energy production (Sterling, 1994). 2,4-DNP is considered primarily as an uncoupler of oxidative phosphorylation, and of proton-coupled fluxes at the plasma membrane and endomembrane via the dissipation of transmembrane electrochemical gradients of protons (Zhan et al., 2010). In the present study, 2,4-DNP inhibited the uptake of PFOS and PFOA by 14.4% and 24.1%, respectively (Fig. 2a), suggesting that the uptake of PFOA and PFOS was an energy-dependent process.



To explore whether the uptake of PFOS and PFOA could be facilitated by water channels or anion channels,  $\text{AgNO}_3$  and glycerol were used as aquaporin inhibitors. Glycerol is used to assay aquaporin activity because it acts as a substrate for aquaporins, and silver blocks the constriction region of the channel by reacting with the sulfhydryl group of a cysteine near the conserved NPA motif (Zhang et al., 2019). A previous study found that the organic anion atrazine was taken up via aquaporins (Su et al., 2005). In the present study, both glycerol and  $\text{AgNO}_3$  significantly inhibited the uptake of PFOS (25.3–30.9%) and PFOA (22.3–33.9%) (Fig. 2a), suggesting that aquaporin served as a channel for the uptake of PFASs by *A. orientale* roots.

The effects of the anion channel blockers 9-AC, DIDS, NPPB on the uptake of PFOS and PFOA were further investigated to understand the uptake mechanism. 9-AC and NPPB block uptake through slow anion channels, while DIDS blocks uptake through rapid anion channels (Barbier-Brygoo et al., 2000; Schroeder et al., 1993). In the present study, the uptake of PFOS was inhibited by 9-AC (11.1%) and NPPB (27.6%), and the uptake of PFOA was inhibited by NPPB (23.9%) and DIDS (16.4%) (Fig. 2b). This suggested that the uptake of PFOS and PFOA by *A. orientale* roots was mediated by anion channels. The competitive effect between PFOS and PFOA (Fig. 2c) further suggested that the uptake of PFOS and PFOA might share a similar transport mechanism.

Water and anion channels in plants are membrane proteins that can transport water and anions depending on the size and charge of their holes (Taiz and Zeiger, 2010). Plant aquaporins represent a large protein family and are divided into many subgroups (e.g., plasma membrane intrinsic proteins, PIPs) (Mudumbi et al., 2017). The available evidence indicates that PFASs have carbon-fluorine bonds (C—F) with a typical size of about 1.35 Å (Krafft and Riess, 2015). This size is smaller than the narrowest pore diameter of aquaporins (approximately 2.8 Å) (Kozono et al., 2002; Mudumbi et al., 2017). Compared with water molecule, anions (e.g.,  $\text{SO}_4^{2-}$ ) are larger (Nightingale, 1959), and the pore diameter of anion channels is larger than that of aquaporins. This suggests that PFASs are likely to be transported by water and anion channels whose pore diameter matches the C—F bond size. In addition, the transpiration water loss showed no significant differences between control group and treatment group (Table S8). This indicated that water and anion channel inhibitors had no effect on the passive diffusion of PFOS and PFOA in roots. Thus, the uptake of PFOS and PFOA in plant roots was an active process that was facilitated by water transporters and anion channels.

#### 4.2. Subcellular distribution

Before a chemical enters the cytoplasm, it must cross the cell walls and the plasmalemma. Cell walls, as the first primary barrier protecting the protoplast against pollutant-induced toxicity, bind organic pollutants and restrict their translocation across cell membranes and combination with cell organelles (Wild et al., 2005; Zhao et al., 2015). Cell organelles which contain lipid contents as high as 15–30%, also showed a great accumulation capacity for lipophilic organic compounds (Chiou et al., 2001; Kang et al., 2010). Thus, cell walls and/or organelles were found to be the dominant storage compartments (>50%) for organic pollutants with high lipophilicity and low solubility (Table S15) (Ju et al., 2019; Kang et al., 2010; Lin et al., 2016). In the present study, only 22.8–40.6% and 5.72–13.2% of PFOA accumulated in cell walls and organelles in *A. orientale*, respectively (Fig. 4 and Table S11). In contrast, most PFOA (46.2–70.8%) accumulated in the soluble fraction (Fig. 4 and Table S11); this was consistent with the distribution of thiamethoxam, which has high water solubility (4100 mg/L) and a low  $\log K_{ow}$  value (−0.13) (Wang et al., 2019). This indicates that anionic PFOA has a different subcellular distribution pattern than lipophilic chemicals. PFOS in roots and leaves (from 2 d to 16 d) was distributed mainly in the cell walls (45.6–58.4%), organelles (14.8–30.7%) and soluble fraction (16.3–36.7%) (Fig. 4 and Table S10). This distribution pattern was consistent with

the distribution of chemicals with low solubility and high lipophilicity (Table S15) (Kang et al., 2010; Zhao et al., 2015).

The distribution patterns of PFOS and PFOA in subcellular components are related to their physicochemical characteristics (e.g., ionic form, solubility, lipophilicity). The composition of cell walls (cellulose embedded in a noncellulosic polysaccharide matrix) varies by species and tissue, but all cell walls bear a negative charge (Miller et al., 2016; Taiz and Zeiger, 2010). Electrostatic repulsion may occur between negatively charged PFOS and PFOA and the negatively charged sites in plant cell walls; this is expected to impede the adsorption of PFOS and PFOA on cell walls and their diffusion through the cell wall matrix. This may help explain the low distribution proportion of PFOS (45.6–58.4%) and PFOA (22.8–40.6%) in cell walls (Table S11). Previous studies reported that organic compounds with higher solubility and low  $\log K_{ow}$  values were distributed more in the water-soluble fraction, while chemicals with higher  $\log K_{ow}$  and lower water solubility accumulated more in the root solid phase (e.g., cell walls, organelles) (Gao et al., 2013; He et al., 2017; Kang et al., 2010; Lin et al., 2016; Wang et al., 2019; Zhao et al., 2016a). Similar results were also found for anionic PFOS and PFOA, which are hydrophobic and oleophobic. PFOA, with higher water solubility and a lower  $\log K$  value (phosphatidylcholine model membrane-water partition coefficient) (Table S1) was distributed more in the water-soluble fraction of roots (46.2–70.8%) than PFOS (22.0–27.4%). PFOS, with a higher  $\log K$  value and lower water solubility (Table S1) accumulated more in cell walls (45.6–47.7%) and organelles (27.0–30.4%) than PFOA (23.1–29.1% in cell walls and 7.6–9.7% in organelles).

The water-soluble fraction consists of the apoplastic zone, the cell matrix between organelles, and the vacuoles (Ju et al., 2019). Chemicals in the apoplastic zone of the root cortex and in vascular bundles can be “dragged” by the waterflow of transpiration stream and transported upward to the aboveground part (Trapp, 2000; Wild et al., 2005). The TEM-EDS results showed that PFOS and PFOA were found in the apoplastic space of both the cortex and the vascular bundle (Fig. S4). This indicated that PFOS and PFOA in plant roots could be transported upward to stems and leaves with the transpiration flow. As the transpiration flow passed through the stem and arrived at the leaf, and the water transpired from the leaf to the atmosphere, PFOS and PFOA in the transpiration stream continuously accumulated in the leaf; thus the leaf acted as a sink for PFASs in the aboveground part (sink function of the leaf) as the uptake time increased. This could help explain the higher concentrations of PFOS and PFOA in leaves than in stems in the uptake experiment (2 d to 16 d) (Fig. S3 and Table S9). PFOS and PFOA in the transpiration stream in stems and leaves could also be absorbed by the cell walls and organelles, as they were in roots. Thus, the uptake process of PFOS and PFOA in roots, stems and leaves is proposed as follows: PFOS and PFOA are first distributed in the soluble fraction (e.g., the apoplastic zone in the roots, the transpiration stream in the stems and leaves). Then a proportion of the PFOS and PFOA in soluble fraction is adsorbed gradually by the cell walls, and a proportion of the PFOS and PFOA in the cell walls passes through the cell walls and plasmalemma and adsorbs onto organelles. A similar uptake process was reported for azoxystrobin in wheat roots (Ju et al., 2019).

A previous study found that the subcellular distribution of chemicals was related to the concentration gradient established between subcellular components (Kang et al., 2010). Due to the sink function of the leaves, the concentration of PFOS in the soluble fraction of leaves was higher than that in stems (2 d–16 d) (Table S9), indicating a higher concentration gradient between the soluble fraction and cell walls in leaves than in stems. Then, more PFOS is distributed in cell walls in leaves than in stems due to the higher concentration gradient. This might result in the differential distribution pattern of PFOS between stems and leaves, in which PFOS was mostly distributed in the cell soluble fraction (43.3–77.7%) of stems and in the cell walls (48.4–57.3%) of leaves (Fig. 4 and Table S11).

#### 4.3. Discharge of PFASs from plant roots to external solution

The dissipation of organic chemicals in plant roots depends on their metabolism, transportation upward to shoots and release into solution (Gao et al., 2013). PFOS and PFOA are difficult to metabolize in plants due to their chemical stability (Ochoa-Herrera et al., 2016). The net upward transportation of PFOS and PFOA from roots to shoots was very low (1.9–16.1% at 2–8 d) in this study (Table S12). The discharge of PFASs from plant roots into the culture solution accounted for 72.9–89.4% and 82.7–85.7% of the dissipation of PFOA and PFOS from roots, respectively. Thus, the dissipation of PFOS and PFOA from plant roots was mainly due to their discharge into the culture solution.

In the present study, most PFOS and PFOA (78.6–82.6%) accumulation in root subcellular components (e.g., cell walls, organelles, and soluble fraction) discharged into the culture solution when the plants were transplanted into the culture solution without PFASs for 16 d (Fig. 5). This indicated that the interaction between PFOS and PFOA and root subcellular components (e.g., cell walls, cell organelles) was not strong and that most PFOS and PFOA (78.6–82.6%) accumulation in root subcellular components was reversible. Meanwhile, the proportions of PFOS and PFOA accumulated in the three components remained constant from 0 d to 16 d (Fig. 5 c and f). This suggests that the efflux of PFASs from the intracellular space (e.g., cell organelles) occurred as easily as that from the extracellular space. However, the mechanisms of PFAS efflux across the plasmalemma still need further research.

#### 4.4. Uptake route

Chemicals dissolved in water enter the plant roots by two routes, i.e., the symplastic route and the apoplastic route (Taiz and Zeiger, 2010). Chemicals on the symplastic route move from cell to cell across plasma membranes or via plasmodesmata, while those on the apoplastic route move through the cell walls and/or intercellular spaces (Roberts and Oparka, 2003). In the cortex of plant *A. orientale*, PFOS and PFOA were found in both the apoplastic zone and symplastic zone after 72 h of exposure (Fig. S4). This indicated that PFOS and PFOA were transported from the external solution to the cortex through both the symplastic route and the apoplastic route. In the vascular bundles of *A. orientale*, PFOS and PFOA were also found in the symplastic zone. This indicated that PFOS and PFOA could be transported from the cortex to the vascular bundle through the symplastic route. It is generally assumed that the apoplastic route from the cortex to the vascular bundle is hindered by the endodermic Casparian strip, which consists of hydrophobic suberin and lignin and is impermeable to water and chemicals (Schreiber et al., 1999). Thus, the chemicals on the apoplastic path are forced to pass through the endodermic plasmalemma. In the present study, PFOS and PFOA were found in the apoplastic zone of the vascular bundle of *A. orientale*, suggesting that the apoplastic route was also involved in the radial transport of PFOS and PFOA from the cortex to the vascular bundle. The symplastic and apoplastic uptake routes of PFOS and PFOA in *A. orientale* were the same as the uptake routes of some pesticides (e.g., atrazine, carbendazim, carboxin and diuron) (Peterson and Edgington, 1975) but different from those of neutral organic contaminants such as phenanthrene, which was mainly located in the apoplastic space in the maize root cortex (Dupuy et al., 2016; Wild et al., 2005).

### 5. Conclusions

This study investigated the uptake mechanism, subcellular distribution, and uptake process of two PFASs by the wetland plant *A. orientale*. PFOA and PFOS could be actively taken up by the roots, and their uptake was facilitated by water and anion channels. PFOA and PFOS were distributed mainly in the water-soluble fraction and in the cell walls, respectively. The uptake process of PFOS and PFOA involves distributing in the soluble fraction, gradually adsorbing to cell walls, passing through

the cell walls and plasmalemma, and then binding with organelles. Both symplastic and apoplastic routes were involved in the radial transport of PFOS and PFOA from the external solution to the vascular bundle of the root. This study provides insight into the uptake mechanism and subcellular distribution characteristics of PFOS and PFOA in wetland plants and promotes the development of mechanism-based phytoremediation strategies.

#### CRediT authorship contribution statement

**Tuan-Tuan Wang:** Investigation, Writing - original draft. **Guang-Guo Ying:** Supervision, Conceptualization, Writing - review & editing. **Liang-Ying He:** Data curation. **You-Sheng Liu:** Formal analysis. **Jian-Liang Zhao:** Formal analysis.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2020.139383>.

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